

jessica.e.snyder@gmail.com

Website: <https://biology.work/>

Orcid: orcid.org/0000-0001-8738-5991

Jessica E. Snyder, PhD

Engineer and explorer seeking interdisciplinary team to build, as much as grow, biotechnology

Work

Universities Space Research Assoc contracted to NASA Ames Research Center

Scientist + Synthetic Biology Task Lead (2017-2020) *Biological industrial design to outfit human missions to space.* Built biotechnology alongside evolutionary and synthetic biologists to reverse engineer life from extreme environments. Projects included: Digitizing a sea sponge to 3D print a living filter, a portable device to make protein-based drugs, and field work in Antarctica.

Massachusetts Institute of Technology, Senseable City Lab

Postdoctoral Fellow (2015-2017) *A somewhat anonymized public health census using biology, data, and civic infrastructure.* As part of the Underworlds team, automated wastewater sampling to measure the bacteria, virus, and chemicals in different neighborhoods. Our geolocated dataset became a heatmap of human health. Instances of the technology were deployed in Boston and Kuwait. The project has become the company BioBot Analytics.

Drexel University, Mechanical Engineering Department

Research Fellow / PhD Candidate (2009-2014) *Invent manufacturing process to build a model of human liver for space flight medicine.* Used scaffold guided tissue engineering to reproduce the anatomical, multi-cell type structure of hepatic tissue. In collaboration with NASA Johnson Space Center, we used the 3D printed model to measure drug metabolism in space-like environments.

Inventions

Heterogeneous filaments, methods of producing the same, scaffolds, methods of producing the same, droplets, ... Sun, Wei, Qudus Hamid, and Jessica Snyder. (2017) <https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2017019300>

Compositions and Methods for Functionalized Patterning of Tissue Engineering Substrates Including Bioprinting Cell-Laden Constructs for Multicompartment Tissue Chambers. Sun, Wei, Jessica Snyder, Robert Chang, Eda Yildirim, Alexander Fridman, and Halim Ayan. U.S. Patent 12 872,992 (2011) <https://www.freepatentsonline.com/y2011/0136162.html>

Service

Montefiore Health System, New York City

Collaborator (2016 - present) *A clinical trial to measure emotion during painful procedures to test the effectiveness of virtual reality distractions.* The goal is to tailor-make immersive entertainment experiences that can replace opiates and guide patients towards calmer emotional states during hospitalization.

Cell Culture Skills: BSL2, human/animal cell lines, microscopy, plate reader assays, bioprinting, bioreactor (rotary, microfluidic, oscillatory compression/relaxation)

Programming Skills: Digital reconstruction of medical images (ImageJ, ITK-SNAP), data analysis (Python, MatLab), computer aided design and manufacturing CAD/CAM (AutoDesk products, SolidWorks, OpenSCAD)

Prototyping Skills: 3D multi-material printing, laser cutting, replica molding (PDMS, gallium), electrospinning, coaxial extrusion, mechanical testing (cyclic uniaxial)

Education

2009 BS/MS Mech Eng, Drexel University

2014 PhD Mech Eng, Drexel University

2015-17 Postdoc Urban Planning, MIT

Adjunct Faculty

2019 Santa Clara University

2021 Cogswell University of Silicon Valley

Affiliations

NIAC - An Astropharmacy, Co-Investigator

iGEM Stanford-Brown Team, Advisor

Blue Marble Space Institute of Science, Affiliate

HISEAS, Analog Astronaut Selene I Mission

World Health Summit, New Voices 2016